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AI Based Cyber Threat Detection and Intelligent Security Assistant

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Abstract:

Cybersecurity is becoming more Challenging due to the wider use of the internet, IoT devices, mobile applications cloud computing and interconnected computer networks has increased the complexity and high frequency of cybersecurity threats. Common attacks such as malware, phishing, Denial-of-Service (DoS), brute-force attacks, port scanning, botnets and unauthorized access are becoming harder to detect using traditional security methods that mainly depend on fixed rules and known attack signatures. To address these challenges, this research proposes an AI -based Cyber Threat Detection and Intelligent System for analysing cybersecurity activities. The Proposed system uses machine learning techniques to examine security related data and identify whether the observed activity is normal or potentially harmful. Along with thread detection, the system also considers unusual pattern so that activities which do not clearly match known attack categories can be identified for future analysis After identifying the threat, the system generates an alert and suggests suitable security measures so that the user can understand the problem and respond accordingly. The performance of the detection models can be measured using accuracy, recall, precision, F1-score and confusion matrix. By combining threat detection, anomaly identification, threat classification and risk assessment, the approach provides a single framework for analysing suspicious activities. It can also reduce the need for continuous manual analysis and provide users with useful information about potential security issues. The system can help users understand suspicious activities at an early stage and get a clearer idea about the detected security issue. It provides information about the type of threat and suitable security measures based on generated alerts. In future, this approach can be improved for real-time monitoring and handling of new types of cyber threats.

Keywords: Anomaly Detection, Risk Assessment, Cybersecurity, Threat Detection, AI, ML Intrusion Detection.

1. Introduction

Nowadays, technology is growing very fast and we have become highly dependent on digital systems. The data of individuals and organizations is stored, processed and exchanged through digital systems. This includes personal information, money, documents and our different online activities. Areas such as cloud computing, online banking, databases and IoT are connected through networks. Due to this connectivity, the risk of cyberattacks is also increasing. Attacks such as phishing, hacking, Dos can lead to data loss or damage, identity misuse, leakage of confidential information, unauthorized access and financial loss (Pascoe et al., 2024). As technology and digitalization are growing, new types of cybercrimes and threats are also appearing. Due to this, the problem of cybersecurity has become extremely acute to ensure the security of digital data and systems (Pascoe et al., 2024). Earlier, cybersecurity systems were based on signature information and rules. Such systems were only able to detect attacks if they had the necessary information about them. Meanwhile, attackers are constantly evolving and changing their attack methods, thereby constantly presenting new challenges and difficulties in their detection and elimination (Buczak & Guven, 2016; Ferrag et al., 2022). Today, computer systems and networks generate a large amount of log data, network traffic and system events. It is difficult to manually check and analyze all this information. This makes it difficult to identify possible attacks from such a large amount of information. So, there is a need for approaches that can analyze this information and identify possible attacks (Buczak & Guven, 2016). Attackers may use various approaches and new methods can also surface. Thus, utilizing a single approach may not suffice to identify attacks. It is here that Artificial Intelligence and Machine Learning can be of use. Through

pattern recognition, the Machine Learning methods can identify anomalous activities or events which might point to a possible attack which would not have been otherwise detected (Buczak & Guven, 2016; Ferrag et al., 2022). Anomaly detection can also help in finding activities that are different from normal behavior Without depending only on rules. These approaches can be helpful when dealing with new and changing attack patterns. This make the system more adaptive and reduces its dependency on manually created security rules (Ferrag et al., 2022). AI and ML can be useful for different cybersecurity problems such as intrusion detection, malware detection, anomaly detection (Buczak & Guven, 2016; Ferrag et al., 2022). However, existing systems often focus on a particular cybersecurity problems and functions into one integrated-systems. Therefore, there is a need for a system that can handle multiple cybersecurity problem together. After studying the existing research, we found that there is a need for a cybersecurity system that does not focus on only one type of threat or only on attack detection. Many existing approaches mainly focus on detecting a particular type attack, but cybersecurity also requires other stages such as threat classification, identification, risk assessment and security recommendations. There is also a need for a system that can work with new or unseen threats may not follow previously known patterns. Therefore, instead of only detecting whether an attack has occurred, the system should also provide useful information about the threat, such as its type and severity and help in understanding what security action can be taken. This shows the need for an integrated cybersecurity framework that can perform multiple stages of threat analysis in one system (Pascoe et al., 2024). To address these problems, we have worked on an AI-based Cyber Threat Detection and intelligent security system. Our system uses AI and Machine learning to analyze cybersecurity data. First, the data is preprocessed and relevant features are extracted from it. After that machine learning techniques are used to detect and classify cyber threats. The system also includes an anomaly detection mechanism that helps in identifying unusual activities and can give an indication of possible attacks. Based on the detected threats, the system performs risk assessment and generates alerts. It also helps in understanding the detected threat and provides intelligent security recommendations about what action can be taken. In this way, the proposed system combines different stages of threat analysis instead of focusing only on detecting on attack. The main objective of this research is to develop an ai based framework that can detect and classify cyber threats, identify abnormal activities and assess the level of risk associated with a threat. The system also generates security alerts and provides security recommendations based on the detected threats. The main focus of this research is to improve automated threat monitoring, so that threats can be automatically analyzed and identified instead of depending mainly on manual monitoring. The research also considers challenges such as false positive alerts, changing attack patterns and previously unseen activities. By considering these challenges, the proposed framework can be further used develop more adaptive, scalable and intelligent cybersecurity systems in the future (Buczak & Guven, 2016; Pascoe et al., 2024).

2.Literature Review

Recent studies have shown the rise in applications of Artificial Intelligence (AI), Machine Learning (ML), Deep Learning (DL), Large Language Models (LLMs) and Natural Language Processing (NLP) in cybersecurity. Researchers have applied these technologies to different tasks such as network threat classification, anomaly detection, malware detection, threat correlation, Cyber Threat Intelligence (CTI) and intelligent security recommendations. AI based systems have the capability of analyzing large Number of security data an identifying pattern that are difficult to detect using traditional techniques. Existing work often focusses on specific attacks while problems such as real-time correlation, scalability, automated response and explainability are still there. These gaps give the AI based Cyber Threat Detection and Intelligent Security Assistant.

2.1. AI and deep learning for cyber threat detection

Ullah et al. (2019) gave a deep learning-based approach to detect cybersecurity threats in Internet of Things (IoT) environments. The main focus of this study is on Malware detection and software piracy detection. Malware files are converted into images and analyzed using a Deep Convolutional Neural Network (DCNN) for malware detection. Source code is processed using tokenization, stemming, LogTF and TF-IDF before analyzing it using a neural network for software piracy detection. This work showed Deep learning can detect useful patterns from security data automatically without depending on manually created features. This work's main focus is on malware and software piracy and does not cover broader real time threat correlation or intelligent security responses. Lee et al. (2019) worked on cyber threat detection method using Artificial Neural Networks (ANNs).

They converted security events into event profiles for identifying possible cyber threats. The method was tested using real world security data along with datasets such as SVM, K-NN, Random Forest, Naive bayes and Decision Tree. The work showed automated threat detection benefited from converting security events into useful feature profiles. The work only focuses on how to detect and classify events and does not include an intelligent assistant that can combine information regarding security and provide recommendations. Wang et al. (2022) worked on an AI powered network threat detection system called AI@NTDS to detect malicious SSH and Telnet activities. They used data collected from Cowrie honeypots and classified attacks using the MITRE ATT&CK Enterprise Tactics framework. The researchers extracted host based,52 message based and geographical features and tested machine learning algorithms such as Randon Forest, K-NN and LightGBM. The best performance was of LightGBM with 99.20% accuracy, 99.75% precision, 99.85% recall and 99.80% F1-score. This work showed that using different types of features can improve network threat detection. The system's main focus is on SSH and Telnet attacks and does not cover other security data such as logs, alerts, vulnerabilities and threat intelligence. Overall, these studies show that deep learning and AI can be used for automated cyber threat detection. Ullah et al. (2019) worked on software piracy and malware, Lee et al. (2019) studied security events and Wang et al. (2022) focused mainly on network attacks. Mostly these methods focus on detecting and classifying threats. They do not combine different security data sources with contextual analysis, threat correlation and intelligent recommendations. This creates a need for a comprehensive system that can detect threats and help users in understanding and responding to them.

Table 1.1: Comparative analysis of AI/ML/DL Based Cyber Threat Detection

Author(s) & Year	Research Focus	Approach / Framework	Results	Limitations
Ullah et al. (2019)	Malware and software piracy detection in IoT	TensorFlow deep neural network, tokenization, TF-IDF, LogTF and Deep Convolutional Neural Network (DCNN) with malware image visualization	The proposed approach demonstrated improved classification performance for malware & piracy detection	Mainly focuses on malware and software piracy in IoT environments and does not address broader real time threat correlation or intelligent response
Lee et al. (2019)	Cyber threat detection using security event profiles	Artificial Neural Network (ANN) using transformed security event profiles; compared with SVM, KNN, Random Forest, Naive Bayes and decision tree	Demonstrated that event profile representation combined with neural networks can support automated cyber threat detection	Primarily focuses on event classification and does not provide integrated contextual threat interpretation or recommendations
Wang et al. (2022)	Network threat detection for Telnet activities and SSH	52 messages based, Cowrie honeypots, host based and geographical features, LightGBM, random Forest and KNN	LightGBM achieved 99.20% accuracy, 99.75% precision, 99.85% recall and 99.80% F1-score	The system is mainly focused on Telnet related attacks and SSH and has limited coverage of heterogenous security events.

Table 1.1 gives a comparative analysis of work that used AI, ML and DL techniques in cyber threat detection. Ullah et al. (2019) worked on malware and software piracy Lee et al. (2019) worked on security event profiles and Wang et al. (2022) worked on Telnet network activities and SSH. These studied demonstrates the effectiveness of Classification done by AI but also shows existing work is specialized towards particular threat categories or environments.

2.2. Intelligent threat correlation and cyber threat intelligence

Hmimou et al. (2025) gave a cybersecurity framework which is a combination of traditional security tools and Large Language Models (LLMs) to connect and detect different cyber threats. The system uses separate agents for tasks such as log analysis, IP-range analysis and email verification. The agents' result is then brought together through central recommendation system. The work uses tools such as ELK, FAISS, Nmap, Suricata, vulnerability databases and LLaMa 3.3-70B with Retrieval Augmented Generation (RAG). The accuracy achieved by the system is 93.6%, improved threat correlation and reduced false positives. The study shows how LLM based agents can combine information from different sources and provide recommendations. Use of multiple agents and tools can make complex systems and increase their computational needs. Spyros et al. (2025) developed an AI based framework that manages Cyber Threat Intelligence (CTI). The system deals with different phases of CTI process, like collecting, analyzing, extracting, enriching, storing, correlating and sharing threat information. It is also a combination of internal sources like IDS/IPS systems, system logs and honeypots and external sources like vulnerability databases and CERT/CSIRT feeds. ML and NLP techniques are used for classification, outlier detection, entity recognition and similarity analysis and tools like Wazuh and MISP are used to manage and share threat intelligence. This work shows that identification of suspicious activities becomes more effective by combining information from different sources but this can make the system complex as well. The work also does not focus on providing a simple assistant that can give explanation regarding the threats to users. Haile et al. (2025) developed a framework for classifying cyber threats and identifying emerging threats. Cyber Threat Intelligence collected from surface, deep and dark web sources is used in the framework. Large amount of cybersecurity information is analyzed by combining ML, DL, NLP and topic modeling techniques. The work covers threats like botnets, malware, spam, phishing, DDOS attacks and ransomware. Algorithms like SVM, Random Forest, XGBoost, LSTM, CNN are evaluated. Identification of changes and trends in cybersecurity discussions are done with LDA and NMF. The study shows that the identification of emerging trends from large amount of unstructured data can be done with the help of AI.

Table 1.2: Comparative Analysis of Intelligent Correlation and Cyber Threat Intelligence

Author(s) and Year	Research Focus	Approach/Framework	Results	Limitations
Hmimou et al. (2025)	Multisource threat detection and correlation	Multi agent architecture integrating LLMs with email verification, IP analysis and log analysis	Achieved 93.6% detection accuracy and showed improved threat correlation and false positive reduction	Multi agent architecture and integration of security tools is increasing system complexity
Spyros et al. (2025)	Holistic Cyber Threat Intelligence management	MISP, ML, NLP, honeypots, vulnerability databases and CTI sources	Supports CTI collection, analysis, extraction, enrichment, storage, correlation and sharing	CTI framework, does not focus on end user intelligent assistant for recommendations
Haile et al. (2025)	Cyber threat classification and detection	ML, NLP, DL and topic modelling using SVM, Naive Bayes, Randon Forest, LSTM, CNN, LDA and NMF	Supports classification of threats like malware, phishing, ransomware and identifies threat patterns	Relies on heterogenous external CTI sources and faces challenges related to scalability, processing and data quality

Table 1.2 presents a comparative analysis that move beyond individual threat classification toward threat correlation and Threat Intelligence. These studies involve architectural and data integration environments.

Many external sources and different techniques can create problems related to data scalability, quality and real time processing. The work focuses mainly on threat classification and detection and it does not provide intelligent security system that is also organization specific. These studies show that cybersecurity is moving from threat detection to threat correlation and contextual analysis. Hmimou et al. (2025) showed how connecting information from different security tools can be done by LLM based agents, Spyros et al. (2025) focused on how to manage CTI process. Haile et al. (2025) showed how identification of emerging threats from large amounts of online information can be done by using AI and NLP. There is a need for a simple and more accessible system that can detect threats and as well as explain them and provide recommendations.

2.3. Threat mitigation and intelligent security assistance using AI

Dhanush Kodi and Thejas (2024) together worked on how to detect and reduce cyber threats that are common in the modern times using AI itself. They studied that new and changing attacks may be difficult for traditional security methods. AI/ML can analyze large volumes of identity suspicious patterns, security data and support automated security tasks. The study highlights the shift from reactive security toward proactive security. These studies can be useful in developing systems that detect threats and also provide security recommendations. The work gives an overview of AI based threat detection and mitigation rather than presenting a complete intelligent security system. Al Siam et al. (2025) worked on how AI can be used in the field of cybersecurity. The work covers techniques like ML, DL, NLP and anomaly detection which are then used for malware detection, intrusion detection, phishing detection, endpoint security and adaptive authentication. Techniques like CNN, SVM, Random Forest and reinforcement learning are also discussed.

Table 1.3: Comparative analysis of AI based Threat Mitigation and Adaptive Cybersecurity

Author(s) and Year	Research Focus	Method / Framework	Results	Limitations
Dhanushkodi and Thejas (2024)	AI based threat detection and mitigation	AI/ML techniques for analyzing huge amount of security operations and information	Identification of predictive threat detection, resource management and automated response as valuable applications of AI in cybersecurity while integrating efficiency	Privacy, integration complexity, Cost, transparency and explainability are some challenges
Alhumam et al.(2025)	Cybersecurity of digital twin environments	Analysis of security threats across component, asset, network of systems and process	Identifies authentication vulnerabilities, data breaches and Dos attacks and unauthorized access, emphasizes continuous monitoring, adaptive security and resilience	Focused on digital twin environments and therefore does not address cyber threat detection system
Al Siam et al.(2025)	Applications and future prospects of AI in cybersecurity	Review of DL, ML, NLP and anomaly detection including CNN, SVM, RNN, Random Forest and reinforcement learning	Identifies applications in intrusion detection, phishing detection, endpoint security, malware detection and authentication	Highlights major problems like false positives/negatives, computational cost, scalability, privacy, explainability rather than providing compete implementation

The work shows AI based Network Intrusion Detection systems can identify unusual network behaviour. Several problems are still there like computational requirements, false positives and false negatives, scalability, privacy and real time adaptability. The study shows that future cybersecurity systems should provide understandable results and adapt to changing attacks. Alhumam et al. (2025) did their research on the problems that can occur regarding cybersecurity in Digital Twin systems. The study reviewed threats at the components, processes, systems, assets and connected networks. The major issues are authentication weaknesses, data breaches, denial of service attacks, unauthorized access and security problems that are caused due to the integration of cloud technologies and IoT. The researchers gave importance to continuous monitoring, authentication, encryption, adaptive security and risk management. They said there is a need for the systems that can adapt in real time, remain resilient and scale effectively. The study showed importance of cybersecurity system that can adapt instead of relying only on fixed security rules. The above studies show needs in modern cybersecurity to move ahead of simple threat detection toward continuous monitoring, mitigation, intelligent assistance and adaptive analysis. Dhanushkodi and Thejas (2024) did their work on threat mitigation that should be proactive in nature, Al Siam et al. (2025) studied the applications of AI in cybersecurity. Alhumam et al. (2025) focused on continuous monitoring and adaptive security. These studies do not fully combine analysis of information from various sources, automated threat detection, threat explanation and user-oriented recommendations in one system.

Table 1.3 compares studies regarding the transition from traditional reactive cybersecurity toward proactive security systems. These studies collectively show that accurate threat detection alone is not sufficient and modern systems require adaptability, explainability and effective mitigation.

2.4 AI, LLMs and practical deployment challenges

Usman et al. (2025) analyzed the use of AI, ML and LLMs in cybersecurity considering energy efficiency and sustainability as well.

Table 1.4: Comparative analysis of AI/Maland LLM Deployment Challenges

Author(s) & Year	Research Focus	Approach / Framework	Results	Limitations
Usman et al. (2025)	AI, ML and LLM applications in cybersecurity with energy efficiency	AI, ML and LLM based threat detection, automated response, model pruning, quantization, edge computing, real time monitoring and accelerator-based architecture	Identifies predictive threat detection, automated response and resource management as important applications of AI in cybersecurity.	Computational cost, complexity, privacy, transparency and explainability are the big challenges
Hmimou et al. 2025)	LLM based threat detection and correlation	Multi agent LLM architecture with RAG and traditional security tools	Demonstrates that LLMs can provide semantic analysis, recommendations and threat correlation	LLM based multi agent architectures require computational and integration resources
Al Siam et al. (2025)	AI based cybersecurity applications and challenges	ML, DL, NLP and anomaly detection approaches	Highlights potential of AI in automated cybersecurity	False Positives/negatives, scalability, privacy etc. remain unresolved
Spyros et al.(2025)	AI based CTI management	MISP, ML, NLP, honeypots, vulnerability databases and CTI sources	Supports CTI collection, analysis, extraction, enrichment, storage, correlation and sharing	CTI framework, does not focus on end user intelligent assistant for recommendations

Applications such as automated response, real time monitoring, predictive threat detection and resource management are discussed in the study. Techniques such as model pruning, edge computing, specialized accelerators and model pruning are used to reduce the resources needed by AI models. Challenges such as computational cost, data privacy, explainability, transparency and system integration are also pointed out. These challenges are considered when developing AI based cybersecurity assistant because AI models can require significant computing resources. A practical system needs a balance between strong threat detection, resource use and understandable results. The study done by Usman et al.(2025) is related to the challenges given by Al Siam et al.(2025). AI and LLMs can improve threat analysis and also increase computational and integration requirements. An effective AI based cybersecurity system should focus on detection accuracy but also computational efficiency, explainability, privacy and practical deployment.

Table 1.4 presents deployment problems regarding AI, ML and LLM based cybersecurity systems. An effective AI based cybersecurity system should focus on detection accuracy but also computational efficiency, explainability, privacy and practical deployment.

3. Research Gap

The literature review shows that there has been significant progress in using AI for cyber threat detection. Many existing studies focus on a particular type of cyber-attack and don not cover multiple cybersecurity areas. This is one of the major research gaps identified from the existing studies. For example, some studies have worked on malware detection while other have focused on network intrusion detection, security event analysis, cyber threat intelligence management and classification of emerging threats. Some recent studies have also started using LLMs and multi agent architecture for threat correlation and contextual security recommendations. However, these advanced approaches are comparatively more complex and may require significant computational resources and complex security infrastructure. To address this research gap, we aim to develop a cybersecurity system and framework that combines AI-based threat detection, threat classification, security event analysis, risk assessment and an intelligent security assistant within a single architecture. The system combines machine learning-based detection with intelligent analysis and security support. The system can give information about the detected threat and help us understand what security action can be taken. In this way, the proposed framework aims to address the limitations identified in the existing research.

4. Proposed Framework

The proposed framework is an AI-Based Cyber Threat Detection and intelligent Security System designed to detect and analyses cyber threats. First cybersecurity data is collected and pre-processed by removing unwanted and duplicate data. Important features are then selected and given to Machine Learning models to identify normal and malicious activities. An anomaly detection module is also used to find unknown activities. When a threat is detected, the system classifies its types and assigns a risk level such as low, medium or high. It then informs the user or security administrator about detected threat.

4.1. Proposed Cyber Threat Detection Framework

The proposed framework provides a simple and systematic way to detect and analyse cyber threats. The system takes cybersecurity data as input, which contains both normal and malicious activities. First the data is pre-processed by handling missing values, duplicate records, unnecessary data and categorical values. After this, important features are selected for analysis. The selected features are given to AI/ML models to identify whether an activity is normal or malicious. When the system finds any unusual activity, it checks what type of threat it could be. The anomaly detection helps the system find activities that are not normal may be related to new or unknown threats. The detected threat is then checked by the risk assessment module to determine its severity. Finally, the system sends an alert and provides suitable security recommendations to help the user take necessary action. The overall framework flow is as shown in Figure 1.1:

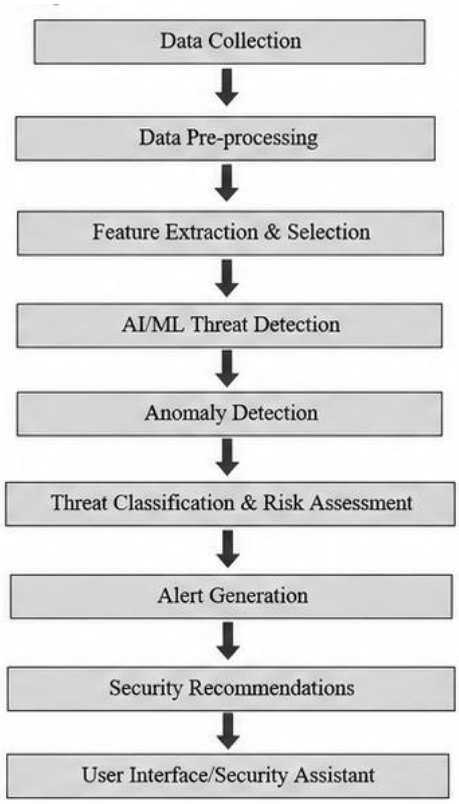


Figure 1.1. Proposed Cyber Threat Detection Framework

4.2. Data collection

The first stage of the proposed framework is data collection. Cybersecurity data is collected form publicly available datasets containing both normal and malicious activities. The datasets may contain information related to network traffic, connections, protocols, connection duration and other security-related features. Different types of cyber threats such as Denial-of-Service, brute force attacks, port scanning, malware-related activities and unauthorized or suspicious activities can be included depending on the selected dataset. The collection data acts as the input for the next stages of the proposed framework (Buczak & Guven, 2016). The mathematical representation is:

$$D = \{(x_i, y_i)\}_{i=1}^{N} \qquad (1)$$

where Drepresents the cybersecurity dataset, $x_i$represents the feature vector of the i^{th}network event, $y_i$represents the corresponding class label and Nrepresents the total number of observations.

4.3. Data pre-processing

The collected cybersecurity data may contain some problems such as missing values, duplicate records, unnecessary information and categorical data. Therefore, the data is cleaned and prepared before applying machine learning algorithm. Firstly the data is cleansed by checking for any missing values and duplicates. Data cleansing consists of the removal of all the superfluous information that may be present but not necessary in the dataset for effective discovery of cyber threats. This information includes categorical data and numerical data. The data from these features has to be converted into coding language understandable by the machine learning model and adjusted if needed. The prepared data is then partitioned into two sets namely: training set and testing set. The training set is used to train the model while the testing set is used to determine the accuracy in predicting and detecting cyber threats (Buczak & Guven, 2016). The mathematical representation of preprocessing:

$$X_{pre} = f_{pre}(X_{raw}) \qquad (2)$$

where X_{raw}represents the raw cybersecurity data, f_{pre}represents the preprocessing function and X_{pre}represents the preprocessed dataset. The prepared dataset is divided into training and testing sets as:

$$D = D_{train} \cup D_{test} \quad (3)$$
$$D_{train} \cap D_{test} = \emptyset \quad (4)$$

where D_{train}represents the training dataset and D_{test}represents the testing dataset.

4.4. Feature extraction and selection

The extracted data is subjected to feature selection. This step is important for feature reduction in cases where the data might have a lot of information. A lot of information may be redundant and not be used in the detection process. Features such as connection information, protocol type, transmission traffic, packet information, conversation time and source and destination information are used to detect cyber threats. Feature selection reduces the amount of data to be processed by eliminating irrelevant information. This process also helps in faster detection of normal and anomalous behavior (Buczak & Guven, 2016).

$$X_{sel} = X[:, S] \quad (5)$$

where Xrepresents the complete feature set, Srepresents the selected subset of relevant features and X_{sel}represents the reduced feature set.

4.5. AI/ML-based threat detection

In this stage, machine learning algorithm are applied to the selected cybersecurity features. The main purpose of this stage is to determine whether a network activity is normal or malicious. Different machine learning algorithms such as Support Vector Machine (SVM), Decision Tree, Random Forest and Logistic Regression can be used to detect and classify cyber threats. Finally, using the training set, the model is trained, which allows it to learn the patterns of both normal and malicious activities. As a result, the obtained model, receiving data about new cybersecurity events as input data, classifies these data as normal or malicious activities. At the same time, in case of detecting malicious activity, using the previously learned data, the model can also classify this activity into one of the previously known categories. Consequently, using such a model, it becomes possible to detect malicious activity automatically and in real-time classify it to identify the attack type (Buczak & Guven, 2016).

$$\hat{y} = f_\theta(x) \quad (6)$$

where $x$represents the selected cybersecurity features, $f_\theta$represents the trained machine learning model, θrepresents the model parameters and $\hat{y}$represents the predicted class. For binary classification:

$$\hat{y} = \begin{cases} 0, & f_\theta(x) < \tau \\ 1, & f_\theta(x) \geq \tau \end{cases} \quad (7)$$

where 0represents normal activity, 1represents malicious activity and τrepresents the classification threshold.

4.6. Anomaly detection

Anomaly detection is an optional stage that allows systems to detect anomalous activities. In other words, using anomaly detection, one can find deviations in the behavior of networks that differ qualitatively from normal activities. As a result, with the help of anomaly detection, it becomes possible to detect not only known attacks but also completely new ones that were not presented in the training data set. The methods for implementing anomaly detection can vary depending on the data set and system requirements; for example, anomaly detection can be implemented using Isolation Forest. Activities classified as anomalies are passed to the analysis and risk assessment stage, where the risk and level of danger of each attack are determined. Anomaly detection helps the system detect suspicious activities that cannot be detected using conventional intrusion detection and protection systems (Liu et al., 2008). For Isolation Forest, the anomaly score can be represented as:

$$s(x) = 2^{-\frac{E(h(x))}{c(n)}} \quad (8)$$

where $E(h(x))$represents the average path length of observation x, $c(n)$represents the normalization factor and $s(x)$represents the anomaly score (Liu et al., 2008). The Isolation Forest method was introduced for isolating anomalous observations through random partitioning of the data. The anomaly decision can be represented as:

$$A(x) = \begin{cases} 1, & s(x) \geq \tau_A \\ 0, & s(x) < \tau_A \end{cases} \quad (9)$$

where $A(x) = 1$indicates anomalous activity, $A(x) = 0$indicates non-anomalous activity and $\tau_A$represents the anomaly threshold.

4.7. Threat classification and risk assessment

The stage of threat classification and risk assessment is necessary to identify the type of attack and assess the level of its danger. To determine the attack type, the system receives the data set of suspicious activity as input data, which is then processed by the previously trained machine learning model. As a result, the machine learning model classifies the data set as normal or malicious activity. In the case of malicious activity, the machine learning model determines which type of attack is present in the data set. Next, using the data from the threat classification, the risk assessment module determines the level of risk and danger of the attack to the system. Finally, based on the threat classification and risk assessment data, the system informs the user of the level of danger and necessary actions to eliminate the threat (Cichonski et al., 2012; Pascoe et al., 2024). NIST guidance describes incident analysis and determining appropriate responses as important components of incident handling, while CSF 2.0 provides a framework for managing cybersecurity risks. The threat classification can be represented as:

$$\hat{c} = \arg \max_{k \in \{1,\dots,K\}} P(y = k \mid x) \qquad (10)$$

where $\hat{c}$represents the predicted threat class, Krepresents the total number of threat categories and $P(y = k \mid x)$represents the probability that input $x$belongs to class k.

The risk level can be represented as:

$$R = P \times I \qquad (11)$$

where Rrepresents the risk level, Prepresents the likelihood of the threat and Irepresents the potential impact of the threat.

4.8. Alert Generation

When the system finds harmful and suspicious activity, it sends and alert to the user or security administrator. The alert provides simple information about the detected activity, including its type, risk level and threat category. This helps the user quickly understand what type of threat has been detected and how serious it may be. The alert generation module helps in quickly information the user about possible security threats. It also reduces the need to manually check a large amount of cybersecurity data and allows faster security action (Cichonski et al., 2012). The alert generation decision can be represented as:

$$Alert(x) = \begin{cases} 1, & R(x) \geq \tau_R \\ 0, & R(x) < \tau_R \end{cases} \qquad (12)$$

where $Alert(x) = 1$indicates that an alert is generated, $R(x)$represents the assessed risk and $\tau_R$represents the predefined risk threshold. Incident-handling guidance includes analyzing incident-related information and determining an appropriate response.

4.9. Security recommendation

The final stage of the proposed framework is the security recommendation module. After detecting the threat and identifying its risk level, the system provides simple security suggestions to help the user respond to the problem. These module makes the system more useful because it not inly detects the threat but also suggestion possible actions to handle it. This can help user take faster and more appropriate security measures (Cichonski et al., 2012; Pascoe et al., 2024). NIST CSF 2.0 provides guidance for organizations to assess, prioritize and communicate cybersecurity efforts and links cybersecurity outcomes to resources and practices. The security recommendation can be represented as:

$$a^* = \arg \max_{a \in A} U(a, \mid cR) \qquad (13)$$

where a^*represents the selected security recommendation, Arepresents the set of possible security actions, $c$represents the identified threat category, Rrepresents the assessed risk level and $U(a, \mid cR)$represents the suitability of action $a$for the identified threat and risk level.

4.10. User interface/security assistant

A User Interface and Security Assistant is the final part of the proposed system. Its main purpose is to provide a simple and easy way for users to interact with the system and understand the results of cyber threat detection. A user-friendly interface can be developed using Streamlit. Through this interface, users can upload cybersecurity data and view the result generated by the system. The interface can show whether the activity is suspicious or normal, the type of threat detected is risk level and the security alert generated by the system. The Security Assistant also provides useful security recommendations. For instance, if the system detects any malicious activity on the network, it could recommend that the user block this activity or review firewall rules or check the network for unauthorized activity. If malicious login activity is detected on the system, it could recommend that they change their passwords or update their account security. The interface works as a communication link between the machine learning model and the user. Instead of only showing a result such as "Attack Detected" the system provides more useful information about the detected activity. The overall process can be represented as shown in Figure 1.2:



Figure 1.2. User Interface/Security Assistant

This makes the system more useful and user-friendly because it not only detects cyber threat but also explains the detects threat and provides possible security actions to the user.

5. Conclusion

The proposed AI-Based Cyber Threat Detection and Intelligence Assistant provides a systematic way to classify, detect and analyses different types of cyber threats. The system includes different steps such as data pre-processing, feature extraction, threat detection, anomaly detection, risk assessment, alert generation and security recommendations. These steps work together to provide a complete approach to cybersecurity monitoring. The article presents the Machine Learning algorithms which are used in the cyber security field and proves that they allow designing a system which does not require the a priori knowledge and can learn by analyzing the security data sets to detect suspicious patterns which are characteristic of potential threats. The anomaly detection part can also help identify unusual activity that may be related to new or unknown threats. Risk assessment and security alerts provides useful information to help users and security teams understand the detected threats and take suitable actions. The proposed system can also reduce manual work and help in analysing a large amount of security data more quickly. However, the performance of the system depends on the quality and variety of the dataset, the machine learning algorithms used and the continuously changing nature of cyber threats.

In the future, the system can be improved by using real-time network data, more cybersecurity data and machine learning methods. It can also provide clearer results, respond automatically to detected threats and new use updated information about new cyber threats. These improvements can help the system detect and handle new and complex cyber threats more effectively.

6. Declaration of interests

The authors of the paper confirm that they have no competing interests that could have affected the work of research, analysis or writing of this research paper. All the authors involved in the process of the research work have reviewed and have given their individual approvals for the final version of the work.

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8. About the Authors

Aishwarya, Chandrika, Lovely Devi and Priya are undergraduate students pursuing a Bachelor of Technology (B.Tech.) in Computer Science and Engineering at Government College Dharamshala, Himachal Pradesh, India. Their academic interests include Machine Learning (M.L), Artificial Intelligence (A.I), cybersecurity and intelligent security systems. The major project "AI-Based Cyber Threat Detection and Intelligent Security Assistance," focusing on using AI and ML techniques to detect cyber threats, classify different types of attacks, assess risk levels, generate security alerts, identify unusual activities and provides useful security recommendations. Through this project, the authors are gaining practical knowledge of cybersecurity and machine learning. Their work includes data pre-processing, feature selection, threat detection, anomaly detection and security analysis. The project has also focus on developing a practical system that can identify suspicious activities and provide suitable security guidance to users. The authors aim to improve their technical and research skills in Artificial Intelligence, cybersecurity and Intelligent Security Assistant. Under the guidance of Dr. Pawan Thakur, Assistant Processor and Head of the Department of B.Tech.(CSE), Government College Dharamshala, the author are improving research skills and learning about cybersecurity, artificial intelligence (AI) and machine learning.

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